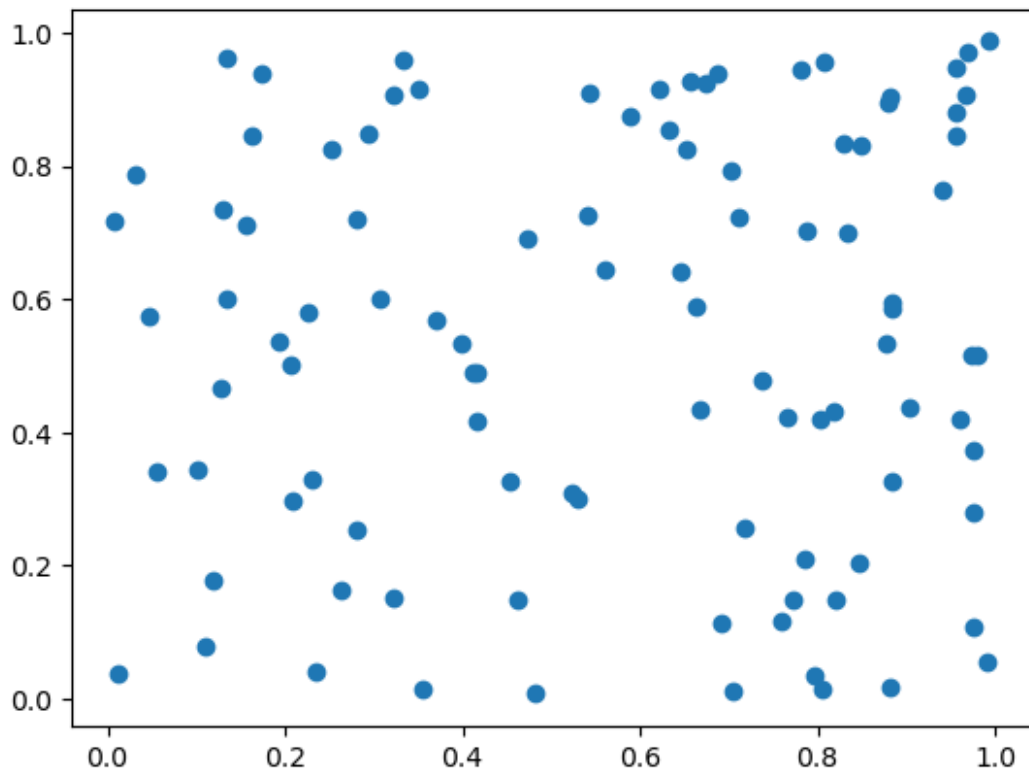


```
import numpy as np
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

X=np.random.rand(100,2)

plt.scatter(X[:,0], X[:,1])
plt.show()
```



```
kmeans=KMeans(n_clusters=3, random_state=0)
kmeans.fit(X)
```

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
warnings.warn(

```
KMeans(n_clusters=3, random_state=0)
```

```
centroids=kmeans.cluster_centers_  
labels = kmeans.labels_
```

```
plt.scatter(X[:, 0], X[:,1], c=labels)  
plt.scatter(centroids[:, 0], centroids[:, 1], marker='x', color='red')
```

```
plt.title("K-Means Clustering")  
plt.show()
```

